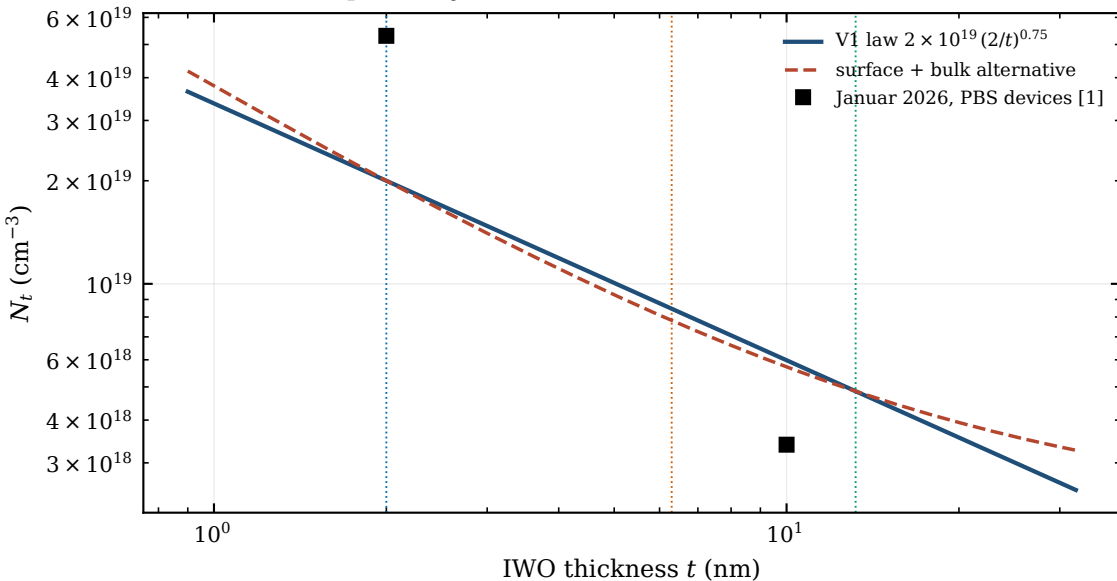


(c) Band-tail trap density $N_t(t)$, model V1



Definition

N_t is the total density of acceptor-like band-tail states below the conduction-band edge. Their density of states is $g_{TA}(E) = N_{TA} \exp[(E - E_c)/W_{TA}]$, so $N_t = \int g_{TA} dE \approx N_{TA} W_{TA}$ with the shared width $W_{TA} = 40 \text{ meV}$ ($T_t = 464 \text{ K}$). ATLAS receives $N_{TA} = N_t/W_{TA}$ (DEFECTS statement). Electrons trapped in these states do not conduct; their filling sets the subthreshold slope and part of the threshold.

Source of the curves

V1 law: FITTED - anchored on the 2 nm ATLAS fit; exponent 0.75 from the $\sim 4\times$ ratio between 13.2 and 2 nm reported in [1]. Surface + bulk alternative: $N_t = N_b + N_s/t$ through the same anchors, $N_b = 2.15 \times 10^{18} \text{ cm}^{-3}$, $N_s = 3.57 \times 10^{12} \text{ cm}^{-2}$ (specialist study S4); it differs by 8 % at 6.3 nm.

References

[1] Januar et al., Small Struct. 7, e202500807 (2026), p. 9 (compact-model extraction on separate PBS devices, unstressed) and Sec. 2.8 / Fig. 5a.